

CONSTRUCTION AND PROJECT MANAGEMENT - I

Section A - Lecture 07

VALUE ENGINEERING

Term-Final Question-to-Slide Mapping | 2016-2024

WHAT IS VALUE ENGINEERING?

- An organized **team effort** aimed at analyzing **Functions** and **Quality** of projects in order to generate practical **cost-effective alternatives** that meet customer requirements.



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Definition of Value Engineering
Lecture slide 4 (PDF page 5)

OBJECTIVE OF VALUE ENGINEERING

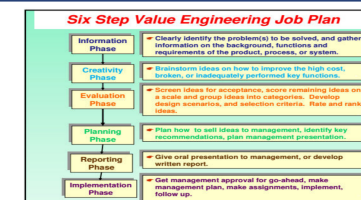
- Improve Project Quality
- Reduce Project Cost
- Foster Innovation
- Eliminate Unnecessary and costly design elements
- Ensure efficient investments in project
- Develop implementation procedures

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Objectives of Value Engineering
Lecture slide 6 (PDF page 7)

VE JOB PLAN

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Six-Step VE Job Plan
Lecture slide 12 (PDF page 13)

Prepared from: 7.0 - Value Engineering.pdf and Construction and Project Management - I Question Bank (2016-2024).

Original exam sections are preserved in the mapping table. The lecture sequence is treated as Section A even when the same topic appeared under Section B in a particular examination.

1. Executive Summary

Mapped questions	Years represented	Mapped marks	Exact matches	Partial matches
8	7 of 9 2017, 2018, 2019, 2020, 2021, 2023, 2024	80	7	1

Core finding: The lecture has strong historical exam relevance. The most repeated pattern is the combined question on the definition of Value Engineering and the six-step VE job plan, appearing in three consecutive examinations (2019-2021).

Coverage by examination year

Year	Coverage
2016	No direct question
2017	Reasons for unnecessary costs
2018	Definition + objectives; reasons for unnecessary costs
2019	Definition + six-step job plan
2020	Definition + job plan
2021	Definition + six-step job plan
2022	No direct question
2023	Definition + objectives
2024	Client value system and project value (partial lecture coverage)

Match rules used

Exact: The lecture directly contains all major elements asked in the question. **Direct:** The lecture clearly supports the answer but may need small explanatory expansion. **Partial:** Only part of the wording or concept is explicitly covered.

2. Detailed Question-to-Slide Mapping

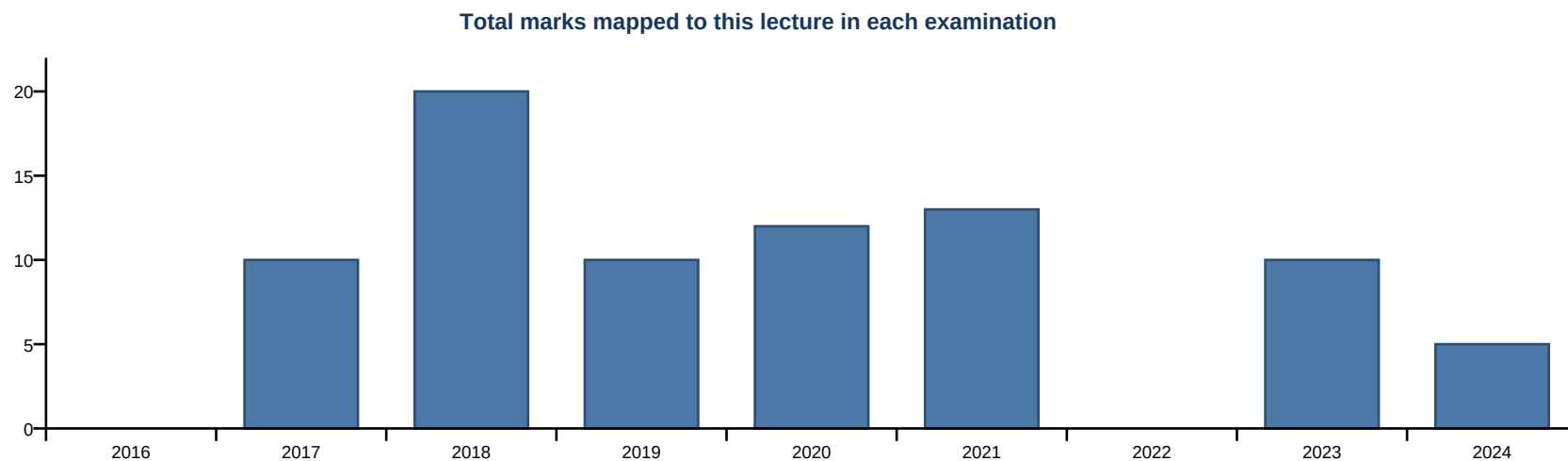
Year	Sec.	Q.	Marks	Question	Lecture slide(s)	PDF page(s)	Match	Coverage note
2017	B	7(c)	10	Concisely explain the reasons for unnecessary costs in a typical construction project.	8-9	9-10	Exact	Both slides list the causes: lack of time/information/ideas, habitual thinking, requirement changes, communication/specification problems, outdated standards, negative attitudes, reluctance to seek advice and poor human relations.
2018	A	2(a)	10	Define value engineering and write down the objectives of value engineering.	4, 6	5, 7	Exact	Slide 4 gives the definition; slide 6 lists six objectives. Slides 1-3 provide useful value, function, quality and life-cycle-cost context.
2018	A	2(b)	10	Briefly describe the reasons for unnecessary costs in a construction project.	8-9	9-10	Exact	Direct repetition of the lecture headings and bullet points.
2019	B	7(a)	10	What is value engineering? Briefly describe the six steps of value engineering job plan in construction sector.	4, 12	5, 13	Exact	Slide 4 defines VE. Slide 12 shows the six phases: Information, Creativity, Evaluation, Planning, Reporting and Implementation.
2020	B	6(a)	12	Define value engineering. Concisely describe the steps of value engineering job plan.	4, 12	5, 13	Exact	The requested definition and job-plan stages are explicitly shown.
2021	B	6(a)	13	What is value engineering? Briefly describe the six steps of value engineering job plan in construction sector.	4, 12	5, 13	Exact	Almost the same wording as 2019; the same two slides form the core answer.
2023	B	7(a)	10	Define value engineering and write down the objectives of value engineering.	4, 6	5, 7	Exact	Same core pattern as 2018 Q2(a).
2024	B	8(a)	5	Define client's value system and explain how it influences the value of a project.	1-3	2-4	Partial	Slides 1-3 define value as Function/Cost and connect value with function, quality and life-cycle cost. The specific term client's value system is not explicitly defined in the lecture.

Question-bank page reference: 2017 p.5; 2018 p.7; 2019 p.10; 2020 p.13; 2021 p.16; 2023 p.21; 2024 p.25 of the combined 25-page file.

3. Repeated Question Patterns

Pattern	Years	Frequency	Priority	Primary slides
Define VE + explain six-step VE job plan	2019, 2020, 2021	3	A+	4 and 12
Define VE + state objectives	2018, 2023	2	A+	4 and 6
Reasons for unnecessary costs	2017, 2018	2	A	8-9
Client value system / value of a project	2024	1	B	1-3 (partial)

Mark trend by year



Interpretation: Value Engineering appeared in seven of the nine examinations. The topic peaked at 20 marks in 2018 because two separate questions were asked.

4. Slide-to-Question Reverse Index

Lecture slide	PDF page	Main content	Linked exam questions	Importance
1	2	Definition of value; Value = Function / Cost	2024 Q8(a) (partial)	B
2	3	Ways to increase value by changing function and/or cost	Supports 2024 Q8(a)	B
3	4	Function, quality and life-cycle cost	Supports 2018 Q2(a), 2024 Q8(a)	A
4	5	Definition of Value Engineering	2018 Q2(a); 2019 Q7(a); 2020 Q6(a); 2021 Q6(a); 2023 Q7(a)	A+
5	6	Why use Value Engineering (visual summary)	Supporting material for objectives	B
6	7	Objectives of Value Engineering	2018 Q2(a); 2023 Q7(a)	A+
7	8	When Value Engineering is used	No direct question in 2016-2024	B
8-9	9-10	Reasons for unnecessary costs	2017 Q7(c); 2018 Q2(b)	A+
10	11	Project life-cycle effort curve	No direct question; supports timing/context	B
11	12	Best time to consider a VE proposal	No direct question; highly examinable	A
12	13	Six-step VE job plan	2019 Q7(a); 2020 Q6(a); 2021 Q6(a)	A+
13	14	Roadblocks/resistance to change	No direct question in 2016-2024	B
14	15	What is not VE?	No direct question; lecture visual is not a complete definition	C

5. Exam-Focused Answer Construction

A. Definition + objectives (2018, 2023)

Use slides 4 and 6. Begin with the lecture definition: Value Engineering is an organized team effort that analyzes project functions and quality to generate practical, cost-effective alternatives meeting customer requirements. Then present the six objectives in separate points: improve quality, reduce cost, foster innovation, eliminate unnecessary/costly design elements, ensure efficient investment and develop implementation procedures.

B. Definition + six-step job plan (2019-2021)

Phase	Lecture-supported focus
1. Information	Identify the problem and gather information on background, functions and requirements.
2. Creativity	Brainstorm ways to improve high-cost, broken or poorly performing functions.
3. Evaluation	Screen, score, rank and develop the strongest ideas.
4. Planning	Prepare recommendations and plan the management presentation.
5. Reporting	Present the proposal orally or through a written report.
6. Implementation	Obtain approval, assign responsibilities, implement and follow up.

C. Reasons for unnecessary costs (2017, 2018)

Use slides 8-9 and group the points for a stronger answer: **information/time deficiencies** (lack of time, knowledge and ideas); **organizational and behavioral causes** (habitual thinking, honest wrong beliefs, negative attitudes, reluctance to seek advice and poor human relations); and **project/document causes** (temporary situations becoming permanent, ownership changes, poor communication/specifications and outdated standards).

D. Client value system (2024)

The lecture supports an answer built around **Value = Function / Cost**, with function, quality and life-cycle cost forming the main value considerations. Explain that the client places different relative importance on required functions, expected quality and total life-cycle cost; these preferences influence design selection and project value.

Important limitation: The uploaded lecture does not explicitly define the term client's value system. The suggested framing above is an inference from slides 1-4 and should be supplemented with the instructor's class notes or prescribed text.

6. High-Priority Revision Plan

Priority	Study target	Slides	Reason
1	Memorize the VE definition and understand each phrase.	4	Required in five examinations.
2	Learn the six job-plan phases in order and add 1-2 lines per phase.	12	Repeated in 2019, 2020 and 2021.
3	Learn all objectives; organize them as quality, cost, innovation and implementation.	6	Asked in 2018 and 2023.
4	Group the reasons for unnecessary costs into logical categories.	8-9	Asked in 2017 and 2018.
5	Understand Value = Function / Cost, plus quality and life-cycle cost.	1-3	Needed for the 2024 value-system question.
6	Review early application, project life cycle and roadblocks.	10-13	Not yet directly asked, but exam-ready content.

Potential questions not yet directly seen in the question bank

1. Explain how value can be increased by changing function and cost.
2. Discuss function, quality and life-cycle cost as issues of value.
3. When should Value Engineering be used?
4. Why is Value Engineering most effective during the conceptual stage?
5. Discuss common roadblocks to implementing Value Engineering proposals.

Overall verdict: Lecture 07 is a high-yield lecture. Preparing slides 4, 6, 8-9 and 12 thoroughly covers almost all historical questions, while slides 1-3 are necessary for the newer 2024 value-system framing.

Quality-control note: Slide numbers refer to the number printed inside the lecture. PDF page numbers count the title page as page 1.